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PI Name:	Midura, Ronald J Ph.D.		
Project Title:	Hypogravity induces vascular atrophy leading to bone loss		
Division Name:	Space Biology		
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Human Research Program Risks:	None		
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Space Biology Special Category:	None		
PI Email:	midurar@ccf.org	**Fax:**	FY 216-444-9198
PI Organization Type:	NON-PROFIT	**Phone:**	216-445-3212
Organization Name:	The Cleveland Clinic Foundation		
PI Address 1:	Biomedical Engineering - ND20		
PI Address 2:	Lerner Research Institute		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44195	**Congressional District:**	11
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
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Task Description:	Hypogravity leads to a loss of ground reaction forces transmitted through the lower appendicular and axial skeleton. Loss of weight bearing (WB) loads leads to a state of progressive osteopenia in these skeletal sites. Hypogravity mediated unloading reduces mechanical compression forces resulting in decreased local bone strains and reductions in hydrodynamic forces within bone tissue. In addition to this mechanism, we hypothesize hypogravity mediates reductions in the vascular supply serving these bone sites that contribute to hypogravity induced osteopenia and helps explain the prolonged time to recover bone mineral density (BMD) and bone strength upon return to Earth. Vasculature associated with bone tissue will be assessed using 3D micro-angiography imaging, which uses contrast agents to fill blood vessels including those in bone followed by micro-CT imaging and histomorphometry. Using micro-angiography, we have developed the tissue and image processing means to assess the bone-associated vascular volumes in the hind limbs of rats. Using the hind limb unloaded (HLU) analog, a time-dependent loss of vascular volume in the lower legs will be monitored and compared to the temporal pattern of progressive bone loss at these skeletal sites. A cause & effect relationship between blood supply and osteopenia will be investigated by directly modulating blood supply to the hind limbs of rats. The blood supply to the hind limbs of WB rats will be reduced using unilateral femoral artery ligation and treatment with a VEGF inhibitor. The blood supply to the hind limbs of HLU rats will be enhanced using a hydrogel-based timed release of VEGF in proximity to the tibia and fibula. Blood flow and vascular volumes will be assessed in these rat groups and a subsequent effect of these treatments on bone volume, BMD, and vascular morphology in the tibiae and fibulae will be determined. A loss of bone-associated blood supply could hamper a proper angiogenic response thereby delaying fracture healing. After HLU exposure, fibulae will be fractured and the extent of repair will be monitored by micro-CT imaging, followed by vascular volume assessments at the fracture callus site using 3D micro-angiography. Vascular volume, bone volume, BMD and vascular morphology within the fracture callus will be compared between HLU and WB rats. Attempts will be made to enhance angiogenesis at the fibular callus site of HLU rats using hydrogel-based timed release of VEGF, while other attempts will decrease angiogenesis at the fibular callus site of WB rats using hydrogel-based timed release of a VEGF inhibitor. This proposal frames a novel hypothesis that could contribute new knowledge impacting on both NASA's Risk of Accelerated Osteoporosis and the risk of Impaired Fracture Healing. If hypogravity unloading leads to an atrophy of bone-associated vasculature, then a negative impact could result contributing to the progressive osteopenia observed during space flight. A reduced bone vascular supply will hamper nutrient and waste product exchange rates in cortical bone thereby stressing osteocytes metabolically, and quite possibly leading to some osteocyte death and localized osteonecrosis. Given the slow remodeling rate of cortical bone, such a possibility could manifest in long term skeletal health complications. First, the presence of any localized osteonecrosis within cortical bone may elevate the risk of fracture upon return to Earth after a long space mission. Second, the remodeling & revascularization of cortical bone in skeletally mature adults takes a long time, and this slow process may help explain why the recovery rates for BMD and bone strength are slow in astronauts returning from long duration space missions. A slow recovery of bone-associated vascular supply may also compound into long term health effects to astronauts in their later years of life (e.g., peripheral artery disease).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our current knowledge base suggests that bone healing is impaired during and after an extended exposure to simulated weightlessness. This study frames a novel hypothesis that states that hypogravity mediated unloading leads to a progressive atrophy of the vasculature suppling lower leg bones. In turn, this regression in bone-associated blood supply and vascular bed volume will have a negative effect on bone tissues resulting in additional and progressive osteopenia beyond that attributed to reductions in local bone strain. Should this association between vascular regression and deficient bone healing hold true, then these studies would manifest relevance to NASA’s existing manned space missions. In addition, these studies may also exhibit importance regarding the underlying causes of impaired bone healing in patients experiencing paralysis, immobility or extended bed rest here on Earth. Finally, this work may also exhibit relevance to other pathologies resulting in the poor healing of recalcitrant bone fractures here on Earth.
	Task Progress Summary - This last August, the Midura lab was afforded the privilege to conduct scientific research upon specimens flown aboard the STS-135 space shuttle mission (CBTM-03 payload). Our portion of work centered on investigating whether vascular alterations occurred in the lower hind limbs of mice flown in space. Due to the relatedness of this spaceflight work to our ground based studies, funding to conduct studies on spaceflight mice was awarded as a supplement to the current award. Thus, spaceflight work will be included in this task book update as section B with the original study task progress as section A. Section C will illuminate tasks progress for the remaining 60 days of year 2 funding. A. Ground-based studies 1) The perfusion protocol has become more efficient with a reduction in sub-par filling of the available vasculature. This is due to standardization of the protocol and consistency in perfusion personnel. While the occasional clotting does occur, the instance of occlusions has diminished sharply for the last two perfusion groups, thereby increasing the quality of our micro-CT acquisitions and providing a more rigorous evaluation of vascular changes. Specifically, deceased rats now remain in the supine position during the entire perfusion process where previously they would be maintained on their right side to facilitate draining of the thoracic and peritoneal cavities during perfusion. Lying on the right side posed an issue for at least several reasons. First, all organs would fall to one side and this could potentially cause blockage of flow where an organ would lie on an afferent or efferent vessel thus hindering flow to extremities or return flow to the heart. Secondly, the first volumes of blood that would expel from the right atrium would rapidly clot due to delayed exposure to heparin as it must pass through the vascular system. These clots would sit within the thoracic cavity adjacent to the hole in the right atrium and would block the flow of perfusate from the heart, increasing vascular pressure. By keeping the rat in the supine position and gently pulling the heart away from the body, these issues are eliminated and clumping of barium within the aorta no longer occurs. 2) Three trials of 20 rats each (60 rats total) have been completed to date. Skeletally mature female retired breeders, aged 7-7.5 months with an average weight of 260 grams, were randomized into one of three groups: a baseline group (used to determine starting parameters for both bone and vasculature), a full weight bearing (WB) group (further subdivided into 4 and 8 week time points), and a hind limb unloaded (HLU) group (further subdivided into 4 and 8 week

Task Progress:	time points). Micro-CT scanning of lower hind limb bones was performed on all animals at time of arrival. Following a 2 week acclimation period, rats in the baseline group (first trial, n=6; second trial, n=4; third trial, n=4) were euthanized and their vasculature was flushed with a 0.9% NaCl/heparin (25U/ml) solution with a constant flow rate of 6 mL/min. Rats were then perfused with a NaCl/barium (median particle size of 1 micron) solution containing 3% gelatin at a constant flow rate of 3 mL/min. Perfusion back pressure was maintained at or below physiological pressure (< 120 mm Hg). Rats in the WB and HLU groups were imaged by microCT during the same time period. The HLU group was then subjected to hind limb unloading for a period of 4 weeks or 8 weeks. Rats in the WB group retained a full weight bearing status for the same time periods. At the end of 4 and 8 weeks, a sampling of WB and HLU rats (first trial: 4 weeks n=3, 8 weeks n=4 each group, second trial: 4 weeks n=4, 8 weeks n=4 each group, third trial: 4 weeks n=4, 8 weeks n=4 each group) were subjected to barium perfusion as described above. 3) Hind limbs from all barium perfused rats were harvested following overnight storage at 4°C to allow gelling of vascular casts. Tissue was fixed in 10% neutral buffered formalin (NBF) and subsequently stored in either 70% ethanol at 4°C (trials one and two) or PBS w/azide (trial 3) until post-perfusion microCT imaging was performed. The storage protocol was revised for trial 3 due to the following: tissue stored in 70% ethanol then processed for paraffin embedding using organic solvents rendered the epitope for the anti-CD31 (Pecam1) antibody undetectable. We have noted that vWF staining was highly variable and decided to migrate to staining of vessels with anti-CD31 in addition to immunohistological localization of smooth muscle actin (SMA). Unfortunately, processing of tissue samples as in trials 1 and 2 limited the utility of the anti-CD31 antibody. Instead, for trial 3 forward, we are using a modified protocol where tissues are fixed in 10% NBF, decalcified in a NBF/ EDTA solution, infiltrated with a 20% sucrose solution and embedded in tissue freezing medium. This process enables cyro-sections that yield effective antibody recognition of the CD31 epitope that will be used for staining of blood vessels. 4) Tissue staining protocols for vWF and SMA have been worked out although variable efficacy of vWF staining has limited its value. The barium-gelatin in-fill may sterically inhibit access to the recognized epitope as unfilled vessels stain brightly while barium filled vessels stain in a punctuate manner or in a greatly reduced fashion. For this reason, tissue from trial 3, as well as samples retrieved from STS-135 (see section B), will be stained for CD31 as described above. A reliable protocol for use of the anti-CD31 antibody for this work will be established before the start of year 3 of this study. Detection of blood vessels for CD31 and SMA will utilize fluorescently labeled secondary antibodies. This type of staining will facilitate quantitation of filled and unfilled vasculature. 5) All animals were injected intraperitoneal with 10 mg/kg alizarin red 14 days- and 10 mg/kgcalcein 2 days- before euthanization. This facilitates dynamic histomophometric analyses of the bone remodeling rates within undecalcified proximal tibiae. 6) While the methodology for analyzing vasculature of barium perfused rats has been decided, modifications to software code and sample processing have been made to increase throughput. Previously, our intentions were to analyze the entire 3 dimensional vascular tree in the lower hind limb (tibial region) of perfused rats using volumes created by micro-CT; however, variable perfusion completeness has caused us to limit our scope to include a region of interest (ROI) that was perfused equally across the majority of samples. This ROI includes the tissue between the tibia and fibula. B. Spaceflight studies 1) Our group was permitted access to sham control mice in both spaceflight and ground control AEMs. This equated to a total of 30 animals (15 spaceflight + 15 ground control) where we utilized the tibial diaphysis of each animals' left hind limb. Upon shuttle landing, the tibial diaphysis of euthanized animals was cut in a transverse manner to separate proximal and distal halves. The proximal end was placed in RNAlater to preserve RNA for genetic analysis of factors associated with endothelial cell alterations. The distal end was placed in 10% NBF. Both pieces were stored at 4°C until use. 2) Tissue from the STS-135 mission stored in RNAlater was separated into two tissue regions, one containing bone and marrow, a second containing skeletal muscle. Muscle tissue was pulverized in the presence of Trizol for RNA extraction. Following RNA recovery by ethanol precipitation, cDNA was made. To date, cDNA created from isolated bone samples was then used to determine gene expression changes using Qiagen gene arrays for endothelial cell function. 3) Tissues from the STS-135 mission stored in 10% NBF have been decalcified, embedded in tissue freezing medium, and cyro-sectioned for immunohistochemical analysis. The staining procedure for CD31 will be finalized soon and staining will commence shortly thereafter. C. Task progress for the remaining 60 days of year 2 Within the next 60 days, all of the RNA samples from the STS-135 mission will be analyzed, both muscle and bone. Immunohistochemical staining of tissue sections from trial 3 and STS-135 samples will soon start and analysis will proceed, as all animal work will be complete. Vascular tree analysis will also commence.
Bibliography Type:	Description: (Last Updated: 03/01/2017)
Articles in Peer-reviewed Journals	Carpenter RD, Lang TF, Bloomfield SA, Bloomberg JJ, Judex S, Keyak JH, Midura RJ, Pajevic PD, Spatz JM. "Effects of long-duration spaceflight, microgravity, and radiation on the neuromuscular, sensorimotor, and skeletal systems." Journal of Cosmology 2010 Oct- Nov;12:3778-80. http://journalofcosmology.com/Mars147.html , Nov-2010
Articles in Peer-reviewed Journals	Androjna C, McCabe NP, Cavanagh PR, Midura RJ. "Effects of spaceflight and skeletal unloading on bone fracture healing." Clinical Reviews in Bone and Mineral Metabolism. 2011. Online First™, 1 February 2011. http://dx.doi.org/10.1007/s12018-011-9080-z , Feb-2011